Using Data and Digital Systems to Improve Trust and Performance in SMEs

Craig McNeile
craig.mcneile@plymouth.ac.uk

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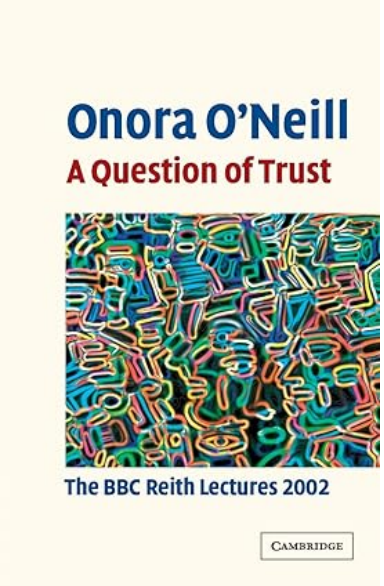
What is trust?



We say we can no longer trust our public services, institutions or the people who run them. The professionals we have to rely on - politicians, doctors, scientists, businessmen and many others - are treated with suspicion. Their word is doubted, their motives questioned.

Quote from the book

> *Confucius told his disciple that three things are needed for government: weapons, food and trust. If a ruler can't hold on to all three, he should give up the weapons first and the food next. Trust should be guarded to the end: without trust we cannot stand.*

Small and Medium-sized Enterprises (SME)

- Devon has nearly 58,000 businesses with over 7,000 new businesses formed in 2020. The majority of Devon's businesses are small to medium enterprises
- The theme of the short talk is that although the cloud services from the US tech companies are useful to SMEs, there may be other sustainable options.

  Large Language Models are too expensive, to the point that anybody funding an "AI startup" is effectively sending that money to Anthropic or OpenAI, who then immediately send that money to Amazon, Google or Microsoft, who are yet to show that they make any profit on selling it.

Ed Zitron `https://www.wheresyoured.at/why-everybody-is-losing-money-on-ai/`

AI Sovereignty

- A critical issue with any AI or machine learning project is the access to data.
- Cloud computing is very convenient for SMEs, but it does require trust in the US big technology companies.

  AI sovereignty is an organization's or nation's capacity to control its artificial intelligence (AI) technology stack, including related IT infrastructure, data, AI models and operations.

Cloud computing

- I once obtained a biggish data set from a large telecom company. The data was for a MSc project. It wasn't customer data, but mostly survey data.
- There had been a number of data breaches of telecom data.
- One of the conditions of obtaining the data set was that it wasn't stored on OneDrive, but the data set was only stored on University storage system.

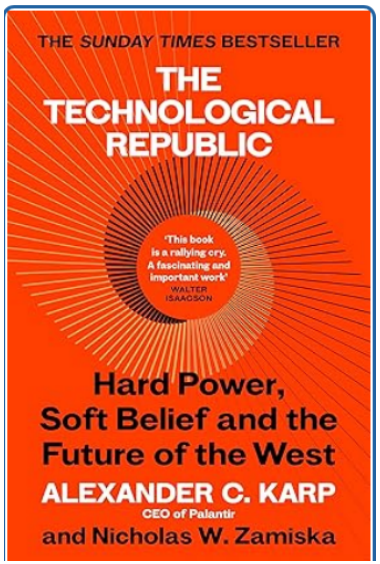
Ensh*ttification / platform decay is a process in which two-sided online products and services decline in quality over time. This is an idea from Cory Doctorow.

Can we trust the US technology companies?

Palantir

Palantir is an American company that develops data integration and analytics platforms.

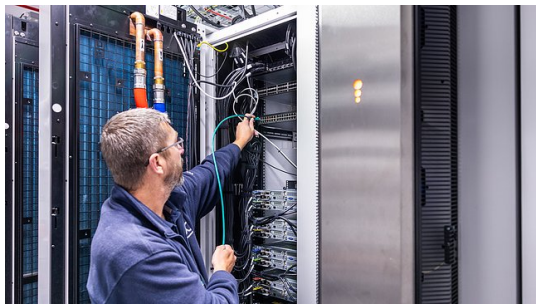
- In 2023 NHS England awarded a £330 million contract to Palantir to provide a new data platform.
- Should we be giving away NHS data to a US company?



- Book by Alexander C. Karp, CEO of Palantir.
- Regain geopolitical power

New HPC system at the University of Plymouth

The University of Plymouth installed a new High Performance Computing (HPC) system at the start of 2025 (investment just over 1 million pounds).



60 × Standard CPU nodes:

- Dual AMD EPYC 9354 (2 × 32 cores)
- 384 GB of RAM per node

2 × High Memory CPU nodes:

- Dual AMD EPYC 9354 (2 × 32 cores)
- 1,536 GB of RAM per node

1 × GPU L40s Accelerated Node:

- Dual AMD EPYC 9354 (2 × 32 cores)
- 1,536 GB of RAM per node
- 8 × NVIDIA L40s PCIe (40 GB per card)

1 × GPU H100 Accelerated Node:

- Dual AMD EPYC 9354 (2 × 32 cores)
- 1,536 GB of RAM per node
- 4 × NVIDIA H100 SXM (80 GB per card)

HPC at the University of Plymouth

Becky Moore • 1st
Driving Commercial Impact Through University–Industry Collaboration
2h •

🚀 Supercharge Your Business with University Supercomputing

Data-driven businesses don't win by working harder — they win by computing smarter.

The University of Plymouth is offering one organisation a fully funded 10-hour project on our Lovelace High-Performance Computing (HPC) system, giving you access to industrial-grade compute normally reserved for global tech giants.

Whether you're training AI models, running complex simulations, or processing massive datasets, this is your chance to unlock capability far beyond a laptop or cloud trial.

What you'll get:
✔️ A tailored Discovery Session with HPC specialists
✔️ A 10-hour proof-of-concept run on Lovelace
✔️ Clear wrap-up insights and recommendations
✔️ Plus: all applicants receive a free 1-hour consultation

- Deadline: Friday 29 May 2026

Isambard-AI facility

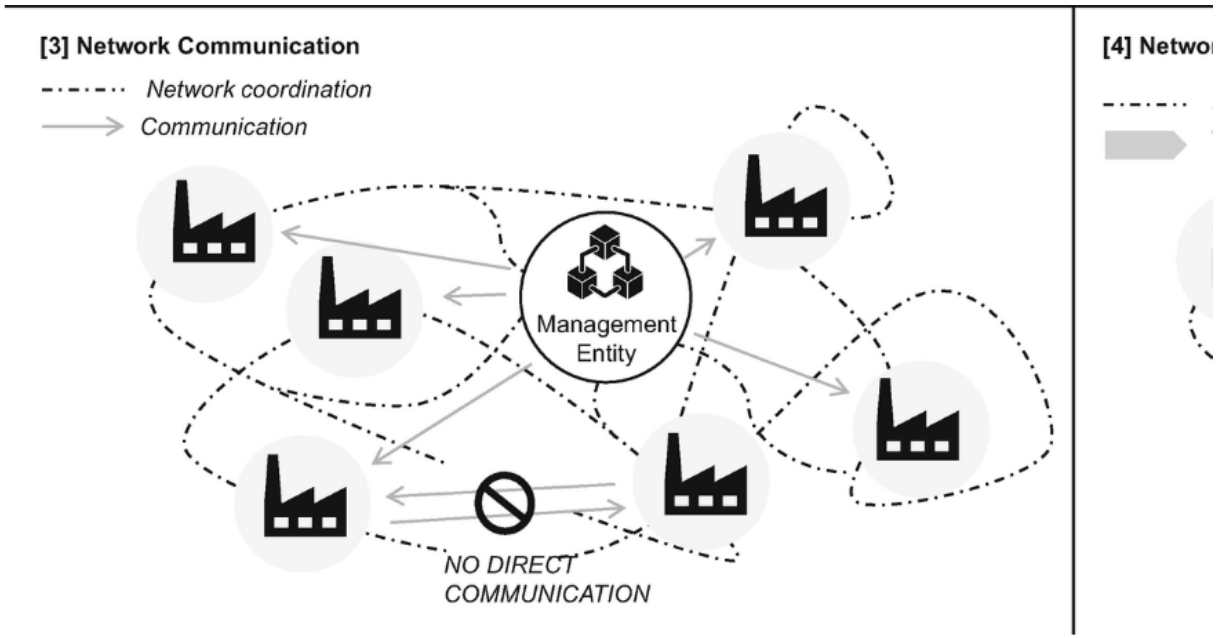
- Isambard-AI facility at the University of Bristol.
- There is a list of the fastest supercomputers in the world.
- Isambard-AI is the 11th fastest system in the world.
- Proposal based application process.

11	**Isambard-AI phase 2** - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cray OS, HPE University of Bristol United Kingdom	1,028,1

Isambard-AI facility

- This modular super-computer is built in a car park.

One way for SMEs to coordinate



Prerequisites for the facilitation of a viable, successful network of local SMEs

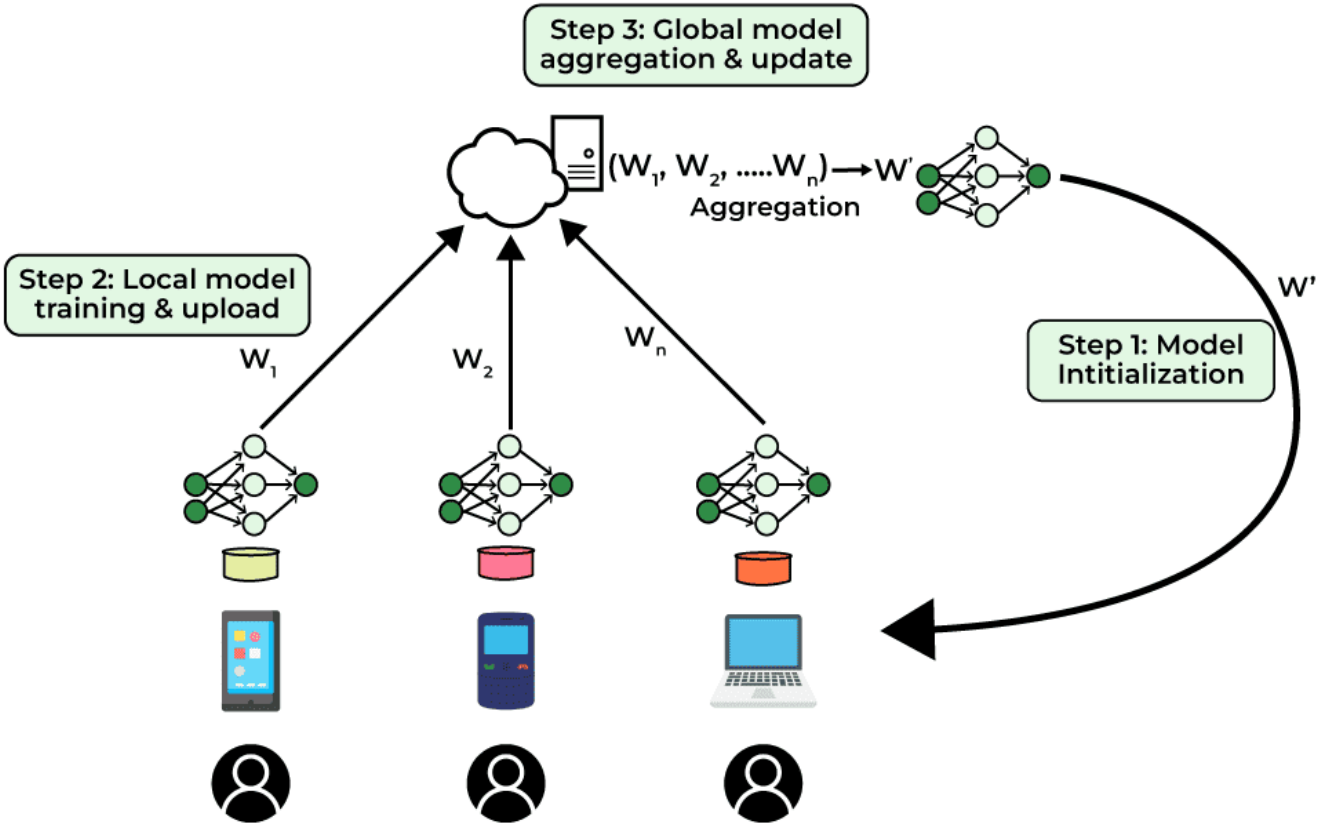
Local Production Networks of SMEs: The Future of Producing Locally? https://link.springer.com/chapter/10.1007/978-3-658-44114-2_21/figures/1

Federated learning

Federated learning is a machine learning technique in a setting where multiple entities (often called clients) collaboratively train a model while keeping their data decentralized.

- Instead of a data set being collected in one storage place, the data stays with the producer.
- This could be data on mobile phones or a network of sensors.
- Multiple organizations, each having their respective trust boundary both in-cloud or on-premise – Multiple organizations (hospitals, banks) come together as a federation to tackle a common machine learning problem (ex: genomics, fraud detection). `https://techcommunity.microsoft.com/blog/azure-ai-foundry-blog/federated-learning-with-azure-machine-learning-powerin` `3824720`
- This requires **trust** that the data is not modified.

Federated learning



Email scam

From: Yinghui Wei <profdesk@wp.pl>
Sent: Monday, June 30, 2025 1:34 PM
To: Craig McNeile <craig.mcneile@plymouth.ac.uk>
Subject: Are you on campus now

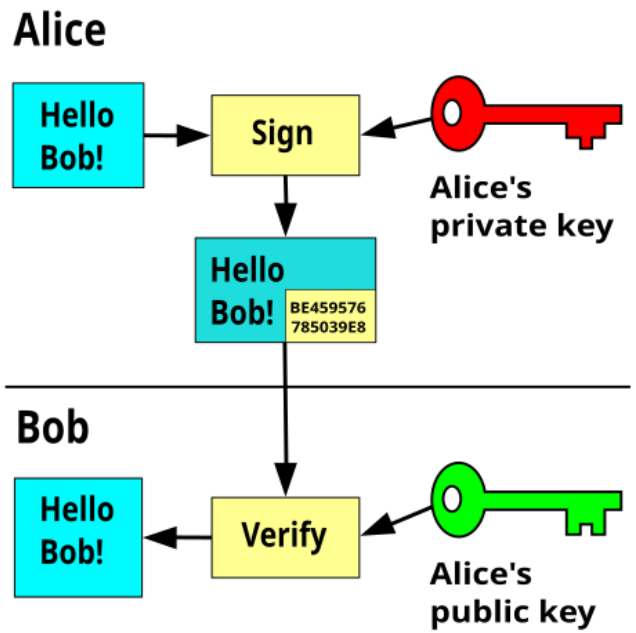
- Occasionally we got emails that claimed to be from someone from the senior management team at the University.
- To my shame, I did answer the email above, because I was in a rush and I didn't notice that the email was not correct.
- I did nothing else, but I know a staff member, who did a video meeting after getting such an email. (I believe the camera was off, so no "deep fake" technology was used.)

How do I trust an email? I do authorize some small expense claims by PhD students via email.

Digital signature

- One way to stop such problems is to use a "digital signature."
- A digital signature is a mathematical scheme for verifying the authenticity of digital messages or documents. A valid digital signature on a message gives a recipient confidence that the message came from a sender known to the recipient.

`https://en.wikipedia.org/wiki/Digital_signature`



Significant Cyber Incidents

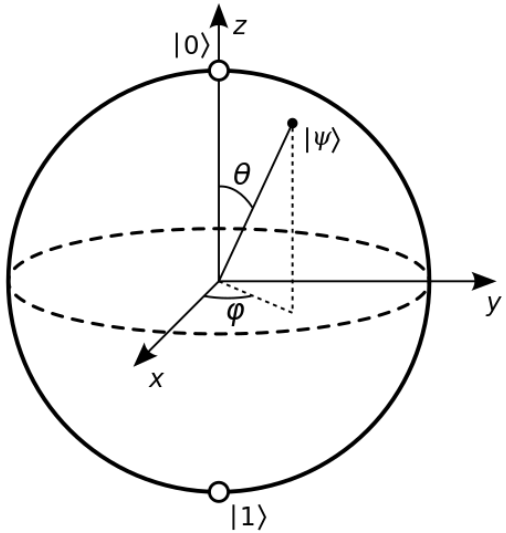
- Canvas software is used to store lecture material and distribute coursework and homework at Colleges and Universities
- Canvas was hacked was hacked last week causing major disruption at thousands of universities and colleges in the USA. **The hackers were paid!**

This is a nightmare for a University.
Unaware, Unfunded, Untrained and Unsupported: A Systematic Review of SME Cybersecurity
`https://arxiv.org/pdf/2309.17186`

- 58% of the attacks registered in 2019 targeted SMEs.
- SMEs were the victims of 43% of breaches.
- 60% of SMEs that suffered a cyberattack go out of business within six months.

Introducing Quantum computers

- The basis of classical computing are bits: **0 and 1**.
- Quantum computers use **qubits** that can be visualized as a sphere.
- Quantum computers use quantum mechanics to solve problems.



https://en.wikipedia.org/wiki/Bloch_sphere

Quantum Computers and encryption

- Current cryptography is based on it being easy to multiply two numbers together.
- It is computationally hard on a classical computer to "factor" a number into two numbers, but computationally easy on a quantum computer.

$$152628 * 627393 = 95757738804$$

$$95757738804 = ??$$

- This is the basis of cryptography. Once a working quantum computer exists it can hack into your encrypted systems.
- Blockchain and crypto will also be at risk from quantum computers.

Quantum computing realities

- Quantum Computers are expensive.
- IBM currently builds Quantum Computers with around 500 qubits but to break cryptography a million qubits are needed, and a lot of technology development is required.
- You to have to trust companies not to sell quantum computers (or time on them) to anyone.

Quantum Computers can not solve all problems efficiently

- Even though Quantum Computers don't have enough qubits to break encryption now, but if criminals get hold of encrypted medical data or messages, they could decode it in 10 years time, when large Quantum Computers are available.
- After worrying you about Quantum Computers breaking encryption.
- Post-quantum cryptography algorithms have been developed that run on standard computers. The issue to protect against Quantum Computers is now making sure your encryption software is updated.
- To help you prepare, Zoom is now offering post-quantum end-to-end encryption in Zoom Workplace.

Uses of Quantum Computers

There are many potential applications of Quantum Computers, beyond breaking encryption.

Optimization supply chains, optimizing telecommunication networks and cargo operations.

Quantum Chemistry Design of drugs and fertilizers.

Quantum Machine Learning potentially able to see new patterns beyond classical machine learning.

PhD projects on quantum computing at the University of Plymouth

- Optimization of layout of turbines in a wind-farm using a quantum computer.
- Quantum Machine learning of pose estimation.

Government's quantum strategy

- Raft of support worth up to £2 billion to establish the UK as a world leader in Quantum, from skills and talent to research and procurement programmes.
- UK aims to become first country in the world to commit to making and deploying Quantum computers at scale by the early 2030s, ushering in a new era for computing which could add £200 billion to the economy by 2045
- A defining technology of the future, Quantum will deliver better drugs and treatments, greener energy, and protect people's finances and national security

Estimates show Quantum could boost productivity by 7 % in the next 2 decades creating more than 100,000 jobs in the process. That would mean £212 billion worth of economic impact.